Odd-characteristic Weil descent onto genus 3

2026-10-14. This is proposal 5 of `problems/ghs-to-smith-pipeline-characteristic-mismatch.md`. It combines a literature reading with an exact native enumeration. It solves no discrete logarithm and claims no cost. Tool: `examples/odd_char_ghs_genus.rs` (std-only Rust). Runs `20261014T1758Z` and `20261014T1800Z` are under `research/odd_char_descent_genus3_20261014/results/`. Protocol `PROTOCOL.md` was frozen at commit `4ad7dd5db`, with one amendment. Tags: [T] theorem (source or derived here), [E] exact computation, [H] heuristic, [O] out of scope or not established. Conductor is not installed on this host. The full report with every quotation is `REPORT.md`.

Answer

- [T, source] **Yes, for $n = 3$.**
  - The curves are $E/\mathbb{F}_{q^3} : y^2 = eg(x)(x-\alpha)(x-\alpha^q)$, with $\alpha \in \mathbb{F}_{q^3} \setminus \mathbb{F}_q$ and $g \in \mathbb{F}_q[x]$ of degree 1 or 2.
  - They have a GHS cover of degree 8 over $\mathbb{P}^1(x)$ by a genus-3 **hyperelliptic** curve over $\mathbb{F}_q$, for every odd q.
  - There are $\Theta(q^2)$ isomorphism classes of such E.
  - Sources: Momose–Chao, ECC 2007, slide 24; Joux–Vitse, Eurocrypt 2012, §4.1, eq. (3), which calls them "the only" such curves, citing Diem 2003, Momose–Chao 2005 and Thériault 2003.
- [E] On all 81 instances ($p \le 31$, three α per p, three choices of h), Joux–Vitse's explicit $H : y^2 = 4N^2h(F/(4N))$ (App. A.1) has $L_{H(T)} = 1 - tT^3 + p^3T^6 = L_{\mathrm{Res}\,E}$. This also matches the repository's own $\mathbb{F}_7$ instance (328; $N = 8, 50, 296$).
- [T, derived] $g = 1 + 2^{m-2}(r-4)$ for every degree-2 x, and the genus is independent of q. Genus 3 needs $(m, r) \in \{(2,6), (3,5)\}$ and a branch-set period of 2 or 3. If $\mathbb{F}_q(j) = \mathbb{F}_{q^n}$ with $n \ge 4$, then every x gives $g \ge 5$.
- [T] The problem record is **not** closed as a proved obstruction, because a family exists. The characteristic-2 pipeline as stated stays blocked: Smith's construction "does not apply in characteristics 2 and 3" (Smith 2009, p. 2).

Sources (exact locations)

- **Chao–Momose, ECC 2007 slides.**
  - sl. 17: Condition (C), "$g = dg_0$, the smallest possible genus".
  - sl. 20: for char $\neq 2$ and even d, covers exist only at $(d, n) = (2, 2)$; for odd d, at $(5, 4)$, at $d = 2^n - 1$, and (decomposable) at $(3, 3)$.
  - sl. 24: the hyperelliptic family.
  - sl. 25: the non-hyperelliptic Types 1 and 2, covering degree 4.
  - sl. 26: when the cover is hyperelliptic.
  - sl. 33: $\#\{\lambda\} \approx q^3/2$.
  - Proofs not read. [T]
- **Joux–Vitse, Eurocrypt 2012 (author PDF).**
  - p. 8, eqs. (3) and (4): $\Theta(q^2)$ curves, all with $4 \mid \#E$. Isogeny-walk conjecture [H].
  - pp. 14–15: the explicit H.
  - p. 16 (App. C): for $n = 6$ and $j \notin \mathbb{F}_{q^3} \cup \mathbb{F}_{q^2}$, "the minimum genus of the Fq-cover obtained by the GHS method is 9". [T]
- **Diem, Compositio 147 (2011).**
  - p. 94: "$g(C) = \sum_L g(L)$, where L runs over all subextensions of k(C)|k(x) of degree two" (Kani–Rosen).
  - Prop. 4.10(a): $g \le (2^n - 1)(2n - 1)$.
  - p. 95: for odd q, $k(E)|k(x)$ is tamely ramified at 4 places.
  - p. 78: Gaudry's $\tilde{O}(q^{2-2/n})$ is heuristic [H].
- **Hess, LMS JCM 7 (2004).**
  - p. 168: the odd-characteristic generalisation is due to Diem.
  - p. 171: "analogous results can be obtained for Kummer extensions".
  - p. 174, Thm 8: the genus is exponential in m (Artin–Schreier). [T]
- **Smith, arXiv:0806.2995.**
  - p. 2: the restriction to characteristic at least 5.
  - Prop. 1 and Def. 1 (pp. 4–5): tractable subgroups.
  - Lemma 1 (p. 6): $\#S(H)$ by factorisation pattern.
  - Props. 2 and 4 (pp. 10, 15): two square conditions.
  - Hypotheses 2–3 [H].
- **Tian, arXiv:2012.07173.**
  - p. 2: prime $n \notin \{3, 5, 7\}$: "the attack fails"; genus 3 at $n = 3$ "necessary condition ... three rational points of order 2".
  - p. 18: (ℓ, ℓ, ℓ)-isogenies $\mathrm{Res}(E) \to \mathrm{Jac}(C)$ for prime order. [T]
- **Abstracts only:** ePrint 2009/236, 2015/805, 2004/073, 2005/119 and 2001/054.
- [O] **Not read**, with no copy reachable from this host:
  - Diem, "The GHS attack in odd characteristic", JRMS 18 (2003);
  - Thériault, JRMS 18 (2003);
  - Diem–Thomé, J. Cryptology (2008);
  - Gaudry, J. Symb. Comput. (2009);
  - the ePrint PDFs (HTTP 403).

Theorems (derived here; proofs in REPORT.md) and enumeration

Set $S = \bigcup_{i<n} \sigma^i B$, $r = |S|$, and $m = \dim \mathrm{span}_{\mathbb{F}_2}\{1_{\sigma^i B}\}$; let k be the period of B.

- **Theorem 1.** $g = 1 + 2^{m-2}(r-4)$, by Riemann–Hurwitz for a tame $(\mathbb{Z}/2)^m$ cover. It equals Diem's subcover sum; [E] checked on every type for $n \le 14$.
- **Theorem 2.**
  - $g = 3 \iff (m, r) \in \{(2,6), (3,5)\}$.
  - $m = 2 \implies k \in \{2, 3\}$, and $(m, r) = (3, 5) \implies k = 3$.
  - $k \ge 4 \implies g \ge 5$.
- **Theorem 3.** For even $n \ge 6$ no type of any period has $g = n$: the genus would force $m = 2$ and $r = n + 3$, but $r \le 8$.
- [E] 601 types for $n \le 8$. Exact branch-set counts for 144 pairs (q, n) ($q = p, p^2$, $p \le 31$) add up to $q^{4n} - q^{2n}$; the completeness check also passes at $q = 5$ for $n \le 14$. 11 unit tests reproduce:
  - Momose–Chao's minimal-cover list;
  - the four $n = 3$ genus-3 shapes;
  - Joux–Vitse's two genus-9 shapes;
  - Diem's bound;
  - Smith's Lemma 1 table.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
min g, period n	1	2	3	5	5	9	7	33	81	49	1793	25	9217	41

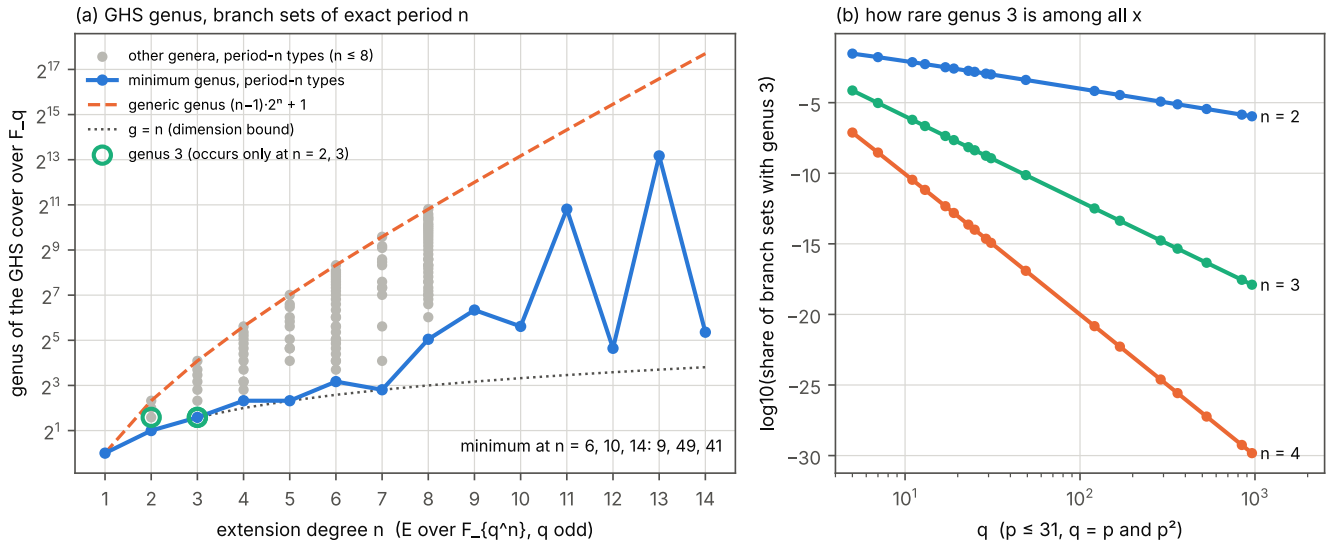


Figure 1: (a) Genera of period-n branch types, the minimum, the generic $(n-1)2^n+1$, and $g=n$; genus 3 is circled and occurs only at $n=2,3$. (b) $\log_{10}$ share of degree-4 branch sets with genus 3. Run `20261014T1800Z`.

The hyperelliptic family and Smith (third arm)

- [E] The Weierstrass cycle type and Smith's count:
  - $(3,3,1,1)$ for $h=x$ and $h=x(x-1)$, so $\#S(H)=3$ rational tractable subgroups;
  - $(6,2)$ for irreducible h, so $\#S(H)=1$;
  - all 81 instances also have $4 \mid \#E(\mathbb{F}_{p^3})$.
- [T, derived] Why: σ normalises the bielliptic group $\{1, \varphi, \varphi^{\sigma}, \varphi^{\sigma^2}\}$, so it acts on each free orbit with exactly one fixed point.
- [O] Smith's Props. 2 and 4 are not checked. It is not checked that J_H/S is a smooth plane-quartic Jacobian. No isogeny is built.
- [T] Types 1 and 2 give plane quartics directly, so Smith matters only for curves whose sole genus-3 cover is hyperelliptic. The census finds 20 such curves at $q=5$ and 42 at $q=7$.
- [O] Conflict with Tian's "three rational points of order 2": with irreducible h, $\#E(\mathbb{F}_{p^3})[2]=2$ on all 27 instances, yet the L-check passes. Not resolved, because the cited sources were not read.

Weil restriction directly ($n=3$)

- [T, standard] With its own polarization, $\mathrm{Res}(E)$ is geometrically $E \times E^{\sigma} \times E^{\sigma^2}$ with the product polarization, so it is never a smooth-curve Jacobian.
- [T, source + E] It is $\mathbb{F}_q$-isogenous to a genus-3 Jacobian
  - via $(2,2,2)$ GHS covers (hyperelliptic family; Types 1 and 2);
  - or via Tian's (ℓ,ℓ,ℓ)-isogenies for prime order.
- [T] The target can be hyperelliptic when $4 \mid \#E$. For prime order, Tian (citing [19]) reports non-hyperelliptic only.
- [O] A full characterisation is not established here.

Genus-k covers, $k=6,10,14$ (`docs/curves/jacobian-transfers.json`)

- [T, source] Joux–Vitse p. 16 give minimum GHS genus 9 at $k=6$.
- [T, source] Momose–Chao sl. 20: no $g=d$ cover for even $d \neq 2$.
- [T + E] By Theorem 3 there is no genus-k type at all. For $\mathbb{F}_p(j)=\mathbb{F}_{p^k}$ the minima are $9, 49, 41$, for every odd p.
- **Plainly:** the 846 `cover_descent` rows priced at $g=k$ use a genus the GHS construction does not deliver.
- [O] Non-GHS covers are not ruled out. The file was not edited.

Does not show

- [O] Costs, logarithms or rho comparisons.
- [O] $\mathbb{F}_q$-models for even n.
- [O] Smith's Props. 2 and 4.
- [O] Characteristic 2 or 3.
- [O] Lean: not formalized, because there is no toolchain on this host.